



Fig. 5: Proposed front panel of the cabinet

lower than 4V. If you are operating above 4V, it is better to use 1.2V mode, as presence of series diodes affects the regulation of the power supply. R3 is used to switch on T1 when current exceeds 600mA through LM317.

Construction and testing

An actual-size, single-side PCB pattern for the 0-50V, 1A variable power supply is shown in Fig. 3 and its component layout in Fig. 4. After assembling all components on the designed PCB, enclose it in a suitable box in such a way that 230V AC mains can be connected easily.

Fix the output terminals on front side of the cabinet along with switches, potmeters and LED1 as shown in Fig. 5. Connect jumper J1 externally on the PCB. Fix switches, LED1 and potmeters VR1 and VR2 with proper labels on the front panel. Verify the test points table before using the circuit.

IC1 and current-boost transistor TIP127/2N6107 should be placed on suitable heat-sinks. S1 is marked as the range-selector switch, while S2 is marked as zero-selection switch. Use 2-pin terminal connectors for output connection. VR1 is labelled for coarse adjustment and VR2 for fine adjustment.

LED1 is used as a power-on indicator. D8, D9 and C2 are used for protection. At 50V, power supply can supply a maximum current of 1A. Choose a good-quality transformer (X1) to support IC1.

Note.

1. Switch S2 to 1.2V when output requirement is above 4V.
2. Switch off AC mains power while switching S1 to high range. Wait for a few seconds until power LED1 goes off.
3. Do not short output terminals.
4. Use proper heat-sinks for IC1 and T1.
5. During testing we used a 20V-0-20V, 2A transformer and output was in the range of 29V, 1.5A to 50V, 220mA when S1 was at position 2 and S2 at 1.2V. Similarly, output was 12V, 800mA to 21V, 100mA when S1 was at position 1 and S2 at 1.2V. **BY**

the three diodes are shorted and actual output of 1.2V is passed on to the output.

Resistor R6 helps series diodes D5 through D7 give better regulation at voltages

STOP GUESSING and know exactly where you should be working with FLIR's patented Infrared Guided Measurement™. IGM identifies and verifies problems invisible to the naked eye to lead you exactly where you should be looking. Electrical to HVAC, water damage to insulation deficiencies, stop guessing and start working with FLIR IGM enhanced meters.

To see the FLIR family of IGM products in action go to FLIR.IN/IGM

IGM

FLIR Systems India Pvt. Ltd.
1111, D Mall, Netaji Subhash Place, Pitampura, New Delhi - 110034
Fax: +91-11-4721 2008 www.flir.in

For more details call us on: +91-11-4580 3555
or write to us at flirindia@flir.com.hk

FLIR | The World's Sixth Sense™

Imagery used for illustration purposes only.



K. Murali Krishna is a former faculty of Aditya Engineering College, Surampalem, Andhra Pradesh. Presently, he is working as telecom technical assistant, BSNL Rajahmundry, Andhra Pradesh